SpaceX Falcon 9

First Stage Landing Prediction

IBM Applied Data Science Capstone Project

Predicting Rocket Reusability to Estimate Launch Cost

90	**4**	**83.3%**	**$62M**
Falcon 9 Launches	ML Models	Prediction Accuracy	Launch Cost

Presented by: Bikash | GitHub: github.com/Bizsubba/IBM-Data-Science-Capstone

Executive Summary

KEY FINDINGS AT A GLANCE

Business Goal

Predict Falcon 9 first-stage landing success to estimate launch cost ($62M vs $165M for competitors)

Data Collected

90 Falcon 9 launches via SpaceX REST API & Wikipedia web scraping, with 17+ features engineered

Best Launch Site

KSC LC-39A achieved the highest success rate at 76.9% — ideal for mission-critical payloads

Best ML Model

Decision Tree had the highest CV accuracy (87.5%). All 4 models achieved 83.3% test accuracy

Overall: Launch success rate improved from 0% (2010) to 90% (2019), demonstrating SpaceX's rapid technological advancement

Introduction

WHY DOES FIRST STAGE RECOVERY MATTER?

The Problem

- SpaceX charges $62M per Falcon 9 launch
- Competitors charge up to $165M per launch
- Cost savings come from reusing the first stage booster
- If we can predict landing success, we can estimate launch cost
- This is valuable intelligence for companies bidding against SpaceX

💰 SpaceX

$62M

per launch

🏢 Others

$165M

per launch

Project Objective

- Collect SpaceX launch data via API & web scraping
- Perform EDA to uncover patterns and trends
- Build ML models to predict landing success
- Identify the best model for deployment

Data Collection Methodology

TWO DATA SOURCES COMBINED

🔗 Source 1: SpaceX REST API

api.spacexdata.com/v4/launches/past

- GET request to SpaceX REST API v4
- 107 total launches retrieved
- Secondary API calls for rocket, payload, launchpad & core details
- Filtered to 90 Falcon 9 launches (removed Falcon 1 & others)
- Missing PayloadMass imputed with column mean

🌐 Source 2: Wikipedia Web Scraping

en.wikipedia.org — Falcon 9 launch history

- Used BeautifulSoup to parse HTML tables
- Extracted 365 total historical launch records
- Captured: date, booster version, launch site, payload, orbit, customer, landing outcome
- Data spans 2010 to 2024
- Saved as spacex_web_scraped.csv for downstream analysis

Combined datasets provide comprehensive coverage of SpaceX's Falcon 9 program from its first flight in 2010

Data Wrangling Methodology

CLEANING & FEATURE ENGINEERING

1 Filter Falcon 9 Only

Removed Falcon 1 and other rocket variants. Retained 90 Falcon 9 launches from 107 total records.

2 Handle Missing Values

Identified missing PayloadMass values. Imputed with column mean to preserve dataset size.

3 Create Landing Outcome Label

Mapped 8 outcome types to binary Class: 1 = successful landing (True ASDS, True RTLS, True Ocean), 0 = failure.

4 One-Hot Encoding

Encoded categorical features: Orbit, LaunchSite, Serial, GridFins, Reused, Legs → 83 total features.

5 Feature Standardization

Applied StandardScaler to normalize all features before ML model training.

6 Train/Test Split

80/20 split with random_state=2: 72 training samples, 18 test samples.

Result: Clean dataset with 90 rows × 83 features | Class distribution: 60 successes (67%) | 30 failures (33%)

EDA & Visual Analytics Methodology

FOUR COMPLEMENTARY ANALYTICAL APPROACHES

📊 EDA with Visualization

Matplotlib · Seaborn

- Scatter plots: Flight No. vs Launch Site
- Bar charts: Success rate by orbit type
- Line chart: Yearly success trend
- Payload mass distributions

🗄️ EDA with SQL

SQLite · Pandas

- Queried launch records using SQL
- Aggregated payload mass statistics
- Filtered by date ranges and outcomes
- Ranked landing outcomes 2010–2017

🗺️ Folium Interactive Maps

Folium · GeoPandas

- Marked all 4 launch sites on world map
- Color-coded success/failure markers
- Measured distances to coastline (0.93 km)
- Proximity analysis to city & highway

📈 Plotly Dash Dashboard

Plotly · Dash

- Interactive dropdown for launch site filter
- Pie chart: success distribution by site
- Scatter plot: payload vs outcome
- Range slider for payload mass filtering

Predictive Analysis Methodology

CLASSIFICATION MODELS WITH HYPERPARAMETER TUNING

📉 Logistic Regression

GridSearchCV (10-fold cross-validation)

Parameters tuned:
C: [0.01, 0.1, 1]
penalty: l2
solver: lbfgs

CV: 84.64%

Test: 83.33%

🎯 Support Vector Machine

GridSearchCV (10-fold cross-validation)

Parameters tuned:
kernel: linear/rbf/poly/sigmoid
C: [0.1, 1, 10]

CV: 84.82%

Test: 83.33%

🌳 Decision Tree

GridSearchCV (10-fold cross-validation)

Parameters tuned:
criterion: gini/entropy
max_depth: 2-10
min_samples: varied

CV: 87.50%

Test: 83.33%

🔍 K-Nearest Neighbors

GridSearchCV (10-fold cross-validation)

Parameters tuned:
n_neighbors: 1-10
algorithm: auto/ball_tree/kd_tree
p: [1, 2]

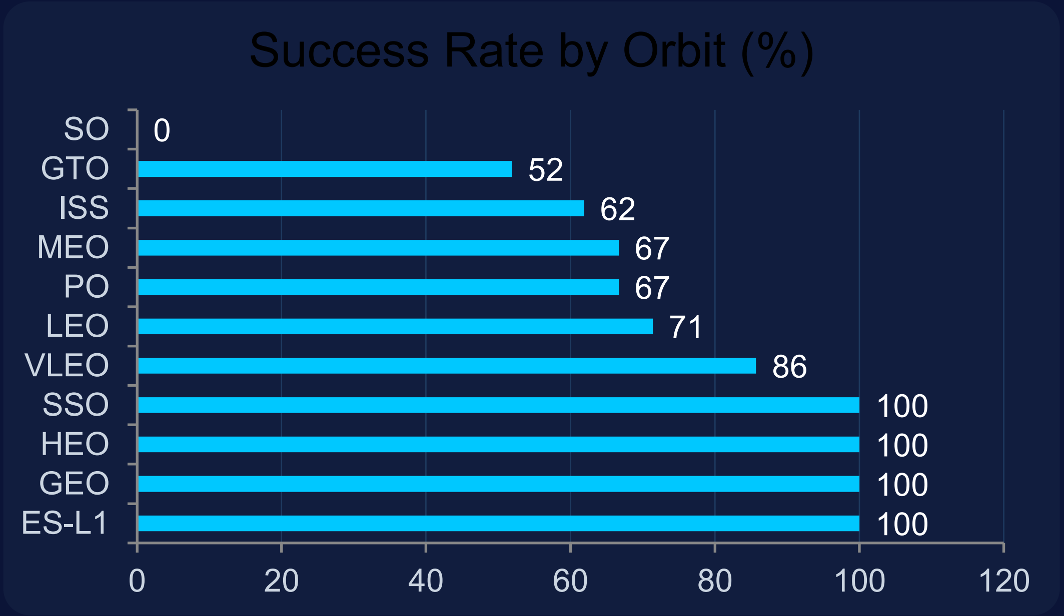
CV: 84.82%

Test: 83.33%

All models evaluated using confusion matrix · Accuracy · 10-fold Cross Validation · GridSearchCV hyperparameter tuning

EDA with Visualization — Results

KEY PATTERNS DISCOVERED



- ✦ ES-L1, GEO, HEO, SSO orbits: 100% success rate
- ✦ SO orbit: 0% success — high-risk missions
- ✦ Success rate grew from 0% (2010) → 90% (2019)
- ✦ Higher flight numbers correlate with higher success

EDA with SQL — Results

KEY QUERIES & FINDINGS

SQL Query 1

Unique Launch Sites

CCAFS SLC-40 · CCAFS LC-40
KSC LC-39A · VAFB SLC-4E

SQL Query 2

Launches from KSC LC-39A

22 total launches

SQL Query 3

First Successful Landing

True RTLS — earliest date in dataset

SQL Query 4

Failed Drone Ship in 2015

2 failures at
CCAFS SLC-40

SQL Query 5

Boosters: Drone Ship 4k-6kg

Multiple F9 FT
boosters identified

SQL Query 6

Top Outcome (2010–2017)

None None: 9
True ASDS: 5

Database: SQLite | Library: Pandas read_sql | Table: SPACEXTBL | Records: 90 Falcon 9 launches

Interactive Map Results — Folium

GEOSPATIAL ANALYSIS OF LAUNCH SITES

Launch Site	Latitude	Longitude	Location
CCAFS LC-40	28.5623°N	80.5774°W	Cape Canaveral, Florida
CCAFS SLC-40	28.5632°N	80.5768°W	Cape Canaveral, Florida
KSC LC-39A	28.5733°N	80.6469°W	Kennedy Space Center, Florida
VAFB SLC-4E	34.6328°N	120.6107°W	Vandenberg, California

📍 Distance to Coastline — 0.93 km

CCAFS SLC-40 is only 0.93 km from the Atlantic coastline — enabling over-ocean trajectories

🟢 Success Markers — Green = Success

Majority of recent launches show green markers, confirming improving success rates over time

🔴 Failure Markers — Red = Failure

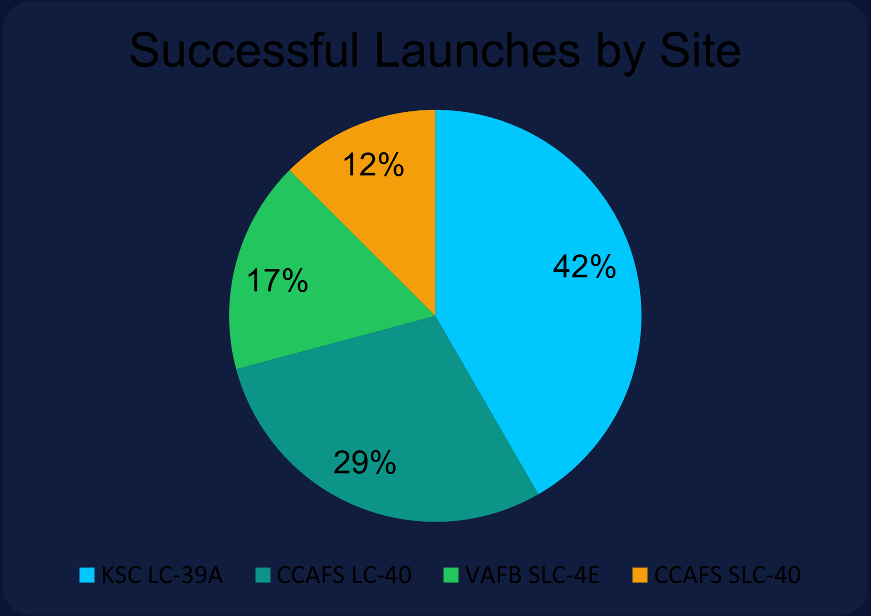
Early launches (2010-2013) predominantly red, showing initial learning curve in rocket recovery

🏖️ Site Proximity — All coastal sites

All US launch sites located near coasts for safety — failed boosters fall into ocean

Plotly Dash Dashboard — Results

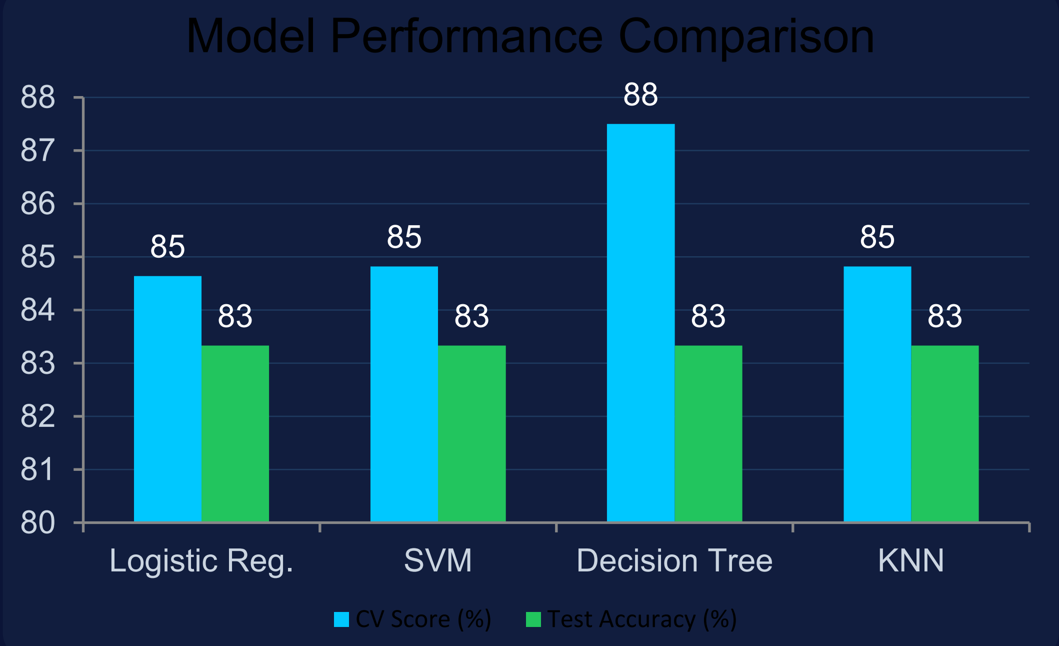
INTERACTIVE LAUNCH ANALYTICS



- 🥇 KSC LC-39A: Best site with 76.9% success rate and 10 successful launches
- 📦 B5 booster version achieved 100% success rate across all payload ranges
- ⚠️ v1.0 booster: 0% success — early technology, no recovery capability
- 🏋️ Payload 2,000–5,500 kg range shows highest success probability

Predictive Analysis — Results

CLASSIFICATION MODEL PERFORMANCE COMPARISON



Logistic Regression

CV: 84.64% | Test: 83.33%

SVM

CV: 84.82% | Test: 83.33%

Decision Tree ⭐

BEST

CV: 87.50% | Test: 83.33%

KNN

CV: 84.82% | Test: 83.33%

🏆 **Winner: Decision Tree — highest CV score of 87.50% with all models achieving equal 83.33% test accuracy**

Conclusion

SUMMARY OF FINDINGS & FUTURE DIRECTIONS

✅ Data Collection

Successfully collected 90 Falcon 9 launches via SpaceX API and 365 historical records via Wikipedia web scraping

📊 EDA Insights

KSC LC-39A is the best launch site (76.9% success). ES-L1, GEO, HEO, SSO orbits achieve 100% success rate

📈 Trend Analysis

Launch success rate grew dramatically from 0% in 2010 to 90% in 2019, reflecting SpaceX's rapid technological maturity

🤖 ML Performance

Decision Tree achieved the best CV score (87.5%). All 4 models tied at 83.3% test accuracy on the held-out set

💡 Business Impact

Accurate landing prediction enables cost estimation: successful reuse = $62M vs $165M for non-reusable competitors

🔭 Future Work

Collect more recent data (2021–2024), try ensemble methods (Random Forest, XGBoost), add weather & payload type features

🔗 GitHub: github.com/Bizsubba/IBM-Data-Science-Capstone | IBM Applied Data Science Capstone — Bikash